

TranslatePsy-AfriSLM: High-Quality Data Scaling For Low-Resource Machine Translation

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Abstract

The rapid progress in Artificial Intelligence has largely bypassed African languages, creating a digital divide that limits AI adoption on the continent. Recent open-source LLMs systematically underperform on African machine translation, while the lack of large-scale, high-quality, open-source parallel data has constrained the development of competitive small language models (SLMs). We introduce **TranslatePsy-AfriSLM**, a collection of open-source MT resources for 19 Sub-Saharan African languages, including curated parallel data, African-specialized synthetic data, and a family of fine-tuned SLMs. Our empirical study shows that unified quality-estimation filtering removes up to 96% of training tokens without degrading quality, and that filtered synthetic data dominates the quality-efficiency Pareto frontier. Fine-tuned on the resulting data mixture, TranslatePsy-AfriSLM outperforms substantially larger systems, including TranslateGemma-27B and Qwen3.5-122B-A10B, with as few as 0.8B parameters.

1 Introduction

The AI underinvestment on the African continent (Nwagbala et al., 2025; Diallo et al., 2025; Isangula, 2025) has created a significant adoption barrier for over a billion people, preventing them from fully exploiting the productivity and collaboration benefits that AI can offer (Maluleke, 2025). A prime example of this is a lack of performant SLMs for Machine Translation (MT), a critical utility for facilitating cross-border communication, trade, and education (Ssemugabi, 2025; Moukatib and Seddik, 2026). However, most frontier open-source LLMs such as Apertus (Hernández-Cano et al., 2025), Qwen3 (Yang et al., 2025), TranslateGemma (Finkelstein et al., 2026), Hunyuan-MT (Zheng et al., 2025) or Qwen3.5 (Team, 2026) systematically underperform on African machine

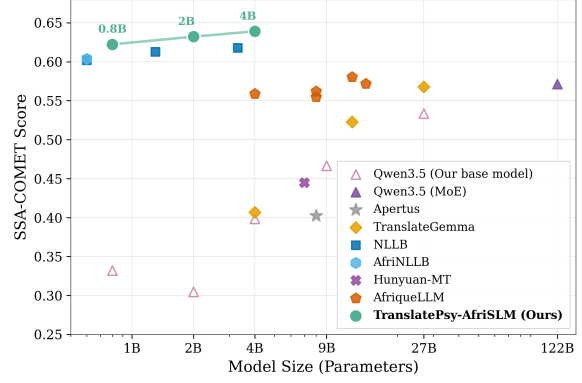


Figure 1: **TranslatePsy-AfriSLM surpasses frontier LLMs (Qwen3.5-122B-A10B, TranslateGemma-27B).** SSA-COMET scores shown on BOUQuET benchmark.

translation (see Figure 1) while incurring substantial running costs due to large parameter counts.

Furthermore, efficiently adapting existing SLMs to support this task remains challenging due to the scarcity of high-quality, open-source data in sufficient quantities, see Background (§2) for a detailed survey. Large internet repositories are highly unstructured and noisy, the training signal is sparse and the token budget dramatically inflated. Existing curated datasets and human-quality translations are too small, offering only modest improvements. Therefore, to support efficient model adaptation for low-resource machine translation, we introduce **TranslatePsy-AfriSLM**¹, an open-source African MT resource comprising high-quality parallel data for 19 Sub-Saharan African languages and a family of highly-capable SLMs, shown in Figure 1. Our approach is systematically validated on key benchmarks, with deep insights and analyses that can guide future research. TranslatePsy-AfriSLM exceeds African translation skills of frontier LLMs such as TranslateGemma-27B and Qwen3.5-27B (and 122B-A10B) with only 0.8B parameters. Our

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¹<https://huggingface.co/collections/qvac/translatepsy-afrislm>

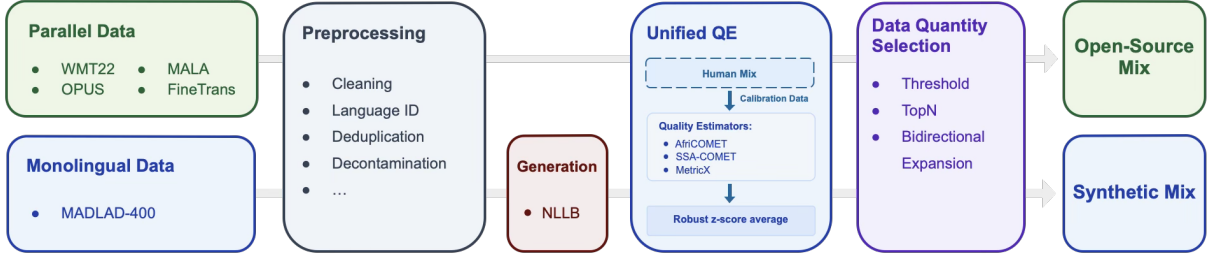


Figure 2: **TranslatePsy-AfriSLM data pipeline.** Parallel and monolingual sources are unified through preprocessing, QE filtering, calibrated on the Human Mix, finally the optimal Open-Source and Synthetic mixes are selected.

smallest model even outperforms the dedicated NLLB encoder-decoder models while preserving conversational capabilities (demo in Figure 13).

2 Background

2.1 African MT Training Resources

Existing African MT datasets reveal a persistent trade-off between scale, coverage, and translation quality. We broadly group them into 3 categories: large but unstructured repositories, medium-sized curated corpora, and small, human-quality datasets.

Large but Unstructured. Data repositories such as OPUS (Tiedemann, 2012), MALA (Ji et al., 2024), WMT22 (Adelani et al., 2022b) and Fine Translations (Penedo et al., 2026) contain large quantities of parallel data, however, the sentence pairs occur in arbitrary sizes, number of languages, translation directions, suffer from high duplication rates, test data contamination and noisy texts. NLLB (Team et al., 2022) leveraged approx. 18B sentence pairs across 200 languages, but did not release a fixed, high-quality corpus suitable for token-efficient adaptation of African MT models.

Medium-Sized and Curated. AfriNLLB² is currently the only readily available, medium-sized, curated dataset for machine translation (Moslem et al., 2026). However, it covers only 9 African languages, and approximately 50% of its ~3M pairs include Arabic or European languages. AfriqueLLM (Yu et al., 2026) adapts frontier LLMs for African languages through continued pretraining. While its general-purpose scope significantly improves MT, its training data was not publicly released (a gap we aim to fill). We report comparisons with AfriqueLLM models in Figure 1 and SLMs trained on AfriNLLB data in Section 5.4.

Small but Human-Quality. Several datasets are available in this subset, e.g. MMT-Africa (Emezue and Dossou, 2021), AfriDOC-MT (Alabi et al., 2025), SMOL (Caswell et al., 2025), LAFAND-MT (Adelani et al., 2022a), WMT24pp (Deutsch et al., 2025), MENYO-MT (Adelani et al., 2021), however, they occur in limited quantities and have uneven language coverage. We show that despite their high-quality translations, such mixtures yield only modest improvements (§5.3), smaller than AfriNLLB and substantially smaller than our best TranslatePsy-AfriSLM mixes. This further motivates our large-scale, quality-focused data curation.

2.2 Quality Estimation for Data Filtering

Afri-COMET (Wang et al., 2024) and more recently, SSA-COMET (Li et al., 2025) were introduced as African-centric, reference-free alternatives to COMET (Rei et al., 2020), COMET-KIWI (Rei et al., 2022b) and MetricX (Juraska et al., 2023). Several prior works have used them to filter parallel text in African languages (Yu et al., 2026; Uemura et al., 2026; Moslem et al., 2026). However, two questions remain unanswered: (1) the comparative usefulness of each metric *as a training data filter*, as opposed to an evaluation metric, validated on large-scale African MT; and (2) whether a combination of individual metrics would perform more consistently, and if so, how to effectively combine multiple QE metrics (§3).

3 TranslatePsy-AfriSLM

We address the heterogeneity of African MT data with a multi-step curation pipeline, shown in Figure 2, applied to open-source and synthetic sentence pairs (§3.1). After standard structural preprocessing, we score raw pairs using Unified Quality Estimation (§3.2), a robust z-score that aggregates multiple QE metrics. We then study data quantity selection methods for post-training (§3.3).

²<https://huggingface.co/datasets/AfriNLP/AfriNLLB-train>

3.1 Data Sources

Parallel Data. We source data from four large, open-source repositories: WMT22 (Adelani et al., 2022b), MALA (Ji et al., 2024), OPUS (Tiedemann, 2012) and Fine Translations (Penedo et al., 2026). Unlike synthetic data, where we can control generation volume and direction, open-source corpora have fixed and often highly uneven distributions across languages and translation directions. As a result, these variables can **vary significantly** between sources. Therefore, we cap each language pair to a maximum of 5M sentence pairs to mitigate ‘winner-takes-all’ effects. This process gives us approximately 427 million raw sentence pairs, see Table 4 for a per-language breakdown.

Monolingual Data. We used the MADLAD-400 corpus (Kudugunta et al., 2023) as a source of data for synthetic data generation via machine translation, see “Synthetic Data Generation” below. For each of the 19 African languages considered in this work, we processed all data available in the corpus. For English, due to the extremely large amounts of data available, we randomly sampled 3.6 million documents. We report detailed, per-language dataset statistics in Table 5.

Preprocessing. This includes standard cleaning, language checks, deduplication, and test set decontamination. Details are provided in Appendix A.1.

Synthetic Data Generation To generate synthetic data, we translated the monolingual data with the NLLB-3.3B encoder-decoder, selected as the teacher model based on our benchmarks in Table 17. See Appendix A.1 for generation settings.

3.2 Unified Quality Estimation

We aim to filter the open-source and synthetic data with AfriCOMET,³ SSA-COMET⁴ and MetricX⁵ QE metrics⁶. As we show later (§5.1), no single quality estimator consistently performs best across all evaluation metrics. This motivates aggregating all three estimators into a single score for more consistent filtering. However, a naïve aggregation seems impractical, as the scores have distinct polarities (higher- versus lower-is-better) and their

ranges vary across QE models, language directions and training corpora. To address this, we unify the quality estimates from AfriCOMET, SSA-COMET and MetricX by mapping them into a shared **robust z-score** (Eq. 2), calibrated on the following dataset.

Human Mix. We compiled ~352K high-quality, human-translated pairs from two datasets (see Appendix A.2 for details) for two distinct purposes: a) to provide a human-quality reference for the robust z-score parameters, and b) to show that high-quality but limited-scale human-translated data alone is insufficient for effective post-training adaptation.

Average Robust z-score. For each translation direction d and for each metric m , we compute Human Mix calibration statistics: the median $\tilde{x}_{d,m}$ and the Median Absolute Deviation (MAD):

$$\text{MAD}_{d,m} = \text{median}(|x_i - \tilde{x}_{d,m}|). \quad (1)$$

Each candidate QE score $x_{i,d,m}$ is then normalized against these statistics:

$$z_{i,d,m} = s_m \cdot \frac{0.6745 (x_{i,d,m} - \tilde{x}_{d,m})}{\text{MAD}_{d,m}} \quad (2)$$

where $s_m \in \{+1, -1\}$ flips lower-is-better metrics such as MetricX, so that higher values always indicate higher quality. By calibrating each score against the same Human Mix statistics (Eq. 2), this normalization places heterogeneous data sources, translation directions, and QE metrics on a common quality scale relative to human-translated data. They can now be aggregated by computing the average of the per-metric normalized scores in Equation 3. We adopt this *average robust z-score* as a unified QE metric, henceforth denoted simply as $\bar{z}$:

$$\bar{z}_{i,d} = \frac{1}{3} \sum_{m=1}^3 z_{i,d,m} \quad (3)$$

This unified score provides a more consistent basis for filtering heterogeneous parallel data.

Directional Filtering. Quality estimation scores are not symmetric with respect to their **argument order**; consequently, for any training pair (X, Y) the score $\bar{z}(X, Y)$ typically differs from $\bar{z}(Y, X)$. The choice of scoring direction therefore determines which sentence pairs pass the filter. We

³<https://huggingface.co/Masakhane/africomet-qe-stl-1.1>

⁴<https://huggingface.co/McGill-NLP/ssa-comet-mt1> - according to Li et al. (2025), this model achieves better results than *ssa-comet-qe*.

⁵<https://huggingface.co/google/metricx-24-hybrid-xl-v2p6withoutreference>.

⁶**Note:** These metrics are used for both quality estimation and reference-based evaluation. We use the names interchangeably depending on local context.

investigate the following strategies for filtering:

- **Aligned:** $\bar{z}(X, Y)$, train $X \rightarrow Y$
- **Reversed:** $\bar{z}(Y, X)$, train $X \rightarrow Y$
- **Mean:** $\frac{1}{2}(\bar{z}(X, Y) + \bar{z}(Y, X))$, train $X \rightarrow Y$

The *aligned* strategy scores each pair in the training direction, whereas the *reversed* strategy scores it in the opposite direction. The *mean* strategy averages the z-scores from both directions, potentially providing a balanced alternative that we also explore.

3.3 Data Quantity Selection

Once the QE strategy is determined, we need to investigate methods of selecting training data quantities: 1) **Threshold** filters out sentence pairs below a given $\bar{z}$ score, 2) **TopN** extends the threshold filter by limiting any single language pair to a maximum of top-N examples to mitigate large imbalances between languages, 3) **Bidirectional expansion** is an orthogonal data augmentation step that reverses each sentence pair $(X, Y) \rightarrow (Y, X)$ to balance translation directions during training.

3.4 Auxiliary Data Mixes

In addition to the open-source and synthetic mixes, we include two auxiliary mixtures to maintain capabilities related to broader machine translation.

Instruct Mix is our multilingual instruction-following dataset with approximately 50% African-language content, totalling 4.6M examples. We include it to preserve general conversational abilities and to extend translation to multi-turn settings. The dataset composition is provided in Appendix A.3.

Asia-Europe Mix covers 38 languages (~ 24 M examples). We include it to preserve translation quality on medium- and high-resource languages and to study whether such data mitigates catastrophic forgetting, see §A.4 for dataset details.

4 Experimental Setup

4.1 Supervised Fine-Tuning (SFT)

We post-train Qwen3.5 (Team, 2026) models using Supervised Fine-Tuning (SFT), as they consistently outperform other general-purpose SLMs on in our preliminary benchmarks. Each model undergoes (full-parameter) fine-tuning for one epoch on the final training mixture selected from our data analyses in §5.2, with loss computed only on assistant tokens. Details are provided in Appendix B.2.

4.2 Evaluation

Language groups. We evaluate MT on 19 languages covered in the TranslatePsy-AfriSLM data mixes, labelled **Africa-IID**⁷, and an additional set of 8 unseen languages, labelled **Africa-OD**⁸, to assess generalisation to out-of-domain languages.

Benchmarks. We evaluate on three translation benchmarks: Flores-200 (Team et al., 2022), BOU-QuET (Team et al., 2026), and Smol (Caswell et al., 2025). Flores-200 is the most widely used benchmark (devtest split, 1,012 sentences), enabling comparison with prior work. BOUQuET is the most recent, linguist-curated multi-way benchmark (test split, 854 sentences) with broad typological coverage. Smol (smolsent split, 863 sentences) provides professionally translated data across 89 languages, used exclusively for evaluation. Each benchmark covers all 19 Africa-IID languages.

Metrics. We adopt four complementary metrics: **COMET-22**⁹ (Rei et al., 2022a), a neural metric trained on human direct assessments, abbreviated to **C22**; **SSA-COMET**¹⁰ (Li et al., 2025), an AfroXLMR-based COMET variant targeting African languages, abbreviated to **SSA**; **MetricX**¹¹ (Juraska et al., 2024), an mT5-based regression metric fine-tuned on DA and MQM ratings, abbreviated to **MX**; and **ChrF++**¹² (Popović, 2017), a character and word-based n-gram F-score robust to morphological variation. COMET-22 and SSA-COMET are reported on a 0–1 scale (higher is better), ChrF++ on a 0–100 scale (higher is better), and MetricX on a 0–25 scale (lower is better).

Baselines We benchmark our SLMs against (1) **General-purpose LLMs:** Qwen3 (Yang et al., 2025) and Qwen3.5 (Team, 2026), two strong general-purpose multilingual LLM families, and Apertus (Hernández-Cano et al., 2025), a massively multilingual model covering 1,800+ languages. Among these, Qwen3.5 achieves the strongest performance, motivating its use as our backbone and enabling a direct measurement of gains from our

⁷ Africa-IID: Afrikaans (afr), Amharic (amh), Hausa (hau), Igbo (ibo), Kinyarwanda (kin), Lingala (lin), Luganda (lug), Malagasy (plt/mlg), Nyanja (nya), Oromo (orm), Nigerian Pidgin (pcm), Shona (sna), Somali (som), Southern Sotho (sot), Swahili (swl/swa), Tswana (tsn), Wolof (wol), Xhosa (xho), Yoruba (yor), Zulu (zul); see https://en.wikipedia.org/wiki/ISO_639-3.

⁸ Africa-OD: Nigerian Pidgin (pcm), Sudanese Arabic (apd), Akan (aka), Tamazight (ber), Kituba (ktu), Bambara (bam), Sepedi (nso) and Mooré (mos).

⁹ <https://huggingface.co/Unbabel/wmt22-comet-da>

¹⁰ <https://huggingface.co/McGill-NLP/ssa-comet-mtl>

¹¹ <https://huggingface.co/google/metricx-24-hybrid-x1-v2p6>

¹² Sacrebleu python library with char_order=6, word_order=2, beta=2, lowercase=False, whitespace=False, eps_smoothing=False

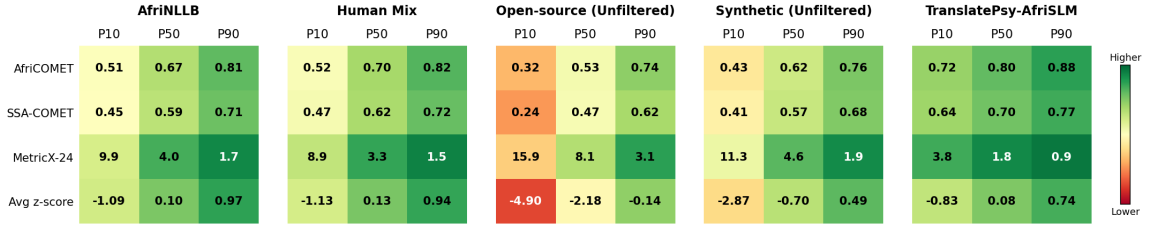


Figure 3: **Comparison of data quality across data sources.** Each column shows the P10, P50, and P90 percentiles for AfriCOMET, SSA-COMET, MetricX, which are normalised into a mean robust z-score (bottom row).

methodology. (2) **Dedicated translation models:** NLLB (Team et al., 2022), an encoder–decoder model trained on large-scale parallel data; AfriNLLB (Moslem et al., 2026), an African-centric variant of NLLB; TranslateGemma (Finkelstein et al., 2026) and Hunyuan-MT (Zheng et al., 2025), decoder-only LLMs post-trained for translation. (3) **African language-specialized models:** AfriqueLLM (Yu et al., 2026), which adapts Gemma-3, Qwen3, and LLaMA-3.1 via continued pretraining on approximately 26B tokens.

5 Results and Analysis

5.1 Quality Estimation: Which is Best?

We first study which quality estimation strategy is most effective for filtering training data. All experiments in this section use English-to-African synthetic data, the only corpus large enough to support controlled comparisons across all 19 language pairs, with over 135 million examples.

No Single Estimator Is Optimal. We compare SSA-COMET, AfriCOMET, and MetricX as individual quality estimators for filtering training data. For each estimator, we score the full training pool in both translation directions, select the top 2 million examples (per language), and use them to fine-tune Qwen3.5-2B. Results on BOUQuET test set are shown in Table 1. Flores-200 and Smol follow the same pattern, as shown in Tables 6 and 7. We observe that each estimator performs best when evaluated by its corresponding evaluation metric. For instance, SSA-COMET estimator consistently achieves the top SSA-COMET score, and the same holds for MetricX. Similarly, the AfriCOMET estimator outperforms MetricX QE when evaluated via COMET variants. Consequently, no single estimator dominates across all metrics. However, the $\bar{z}$ score does provide a balanced alternative, achieving **at least the second best score across all metrics**. We therefore adopt it as our QE metric.

Quality estimator	C22	SSA	MX	ChrF++
eng-xx (English → African language(s))				
z-score	0.765	0.643	3.80	50.1
SSA-COMET	0.766	0.647	3.96	50.5
AfriCOMET	0.763	0.637	4.03	49.8
MetricX	0.760	0.632	3.62	49.6
xx-eng (African language(s) → English)				
z-score	0.786	0.606	4.59	52.7
SSA-COMET	0.785	0.608	4.82	53.2
AfriCOMET	0.783	0.602	4.71	52.4
MetricX	0.782	0.598	4.50	52.1

Table 1: **Translation quality with individual versus unified QE metrics** over BOUQuET test set.

Aligned QE Is Best. We evaluate whether QE filtering should be applied in the training direction (aligned), in the opposite of training direction (reversed) or both (mean). This issue arises in back-translation workflows, where synthetic pairs may be generated and filtered in one direction, then reversed to form training examples in the opposite direction. In such cases, the QE scoring direction becomes *reversed* relative to the final training direction. To test this, we compare otherwise identical models filtered using the *aligned*, *reversed*, and *mean* strategies defined in §3.2. Results in Table 2, averaged across translation directions and test sets, show that reversed filtering substantially degrades performance, especially on MetricX (−12.0%) and SSA-COMET (−3.1%), while the mean strategy is closer but still generally below the aligned reference. This indicates that QE filtering **should be aligned with the final training direction**, even when synthetic data is generated through back-translation. We thus use aligned QE filtering for the remainder of the paper.

5.2 How to Select Data?

With unified QE established, we study how training performance changes as a function of data quality and quantity. We sweep the z-score thresholds and

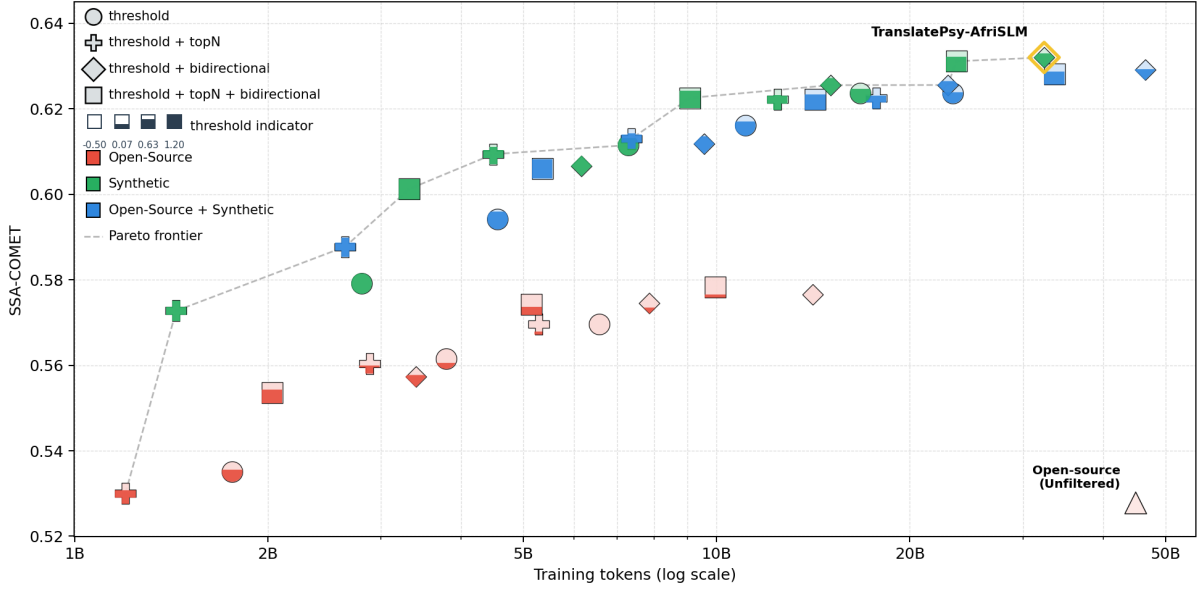


Figure 4: **The Pareto frontier of performance (SSA-COMET) versus training tokens on BOUQuET.** Each shape is a fine-tuned model configuration. Colours indicate data source, shapes indicate the filtering strategy, and the shape fill indicates the threshold (the fuller the shape, the higher the threshold). Detailed plots for each metric and each dataset are shown in Figure 12. The full training budgets and threshold details are reported in Table 15.

Direction	C22	SSA	MX	ChrF++
Aligned (Ref.)	0	0	0	0
Mean	-0.4%	-0.9%	-3.9%	+0.7%
Reversed	-1.5%	-3.1%	-12.0%	+0.1%

Table 2: **Training versus QE scoring directions.** Percentages show average differences to *aligned* reference.

combine them with the three strategies introduced in §3.3 across open-source, synthetic and combined data sources. For each mix, we fine-tune Qwen3.5-2B under the same recipe. Our key observations:

1. **Quality filtering is highly effective.** Figure 4 (bottom right) shows the “Open-source (Unfiltered)” model trained only on raw sentence pairs, totalling 44.93B tokens. Its lower relative quality can clearly be seen in Figure 3. In contrast, a filtered configuration (bottom left) reaches a comparable SSA-COMET score (0.530 vs. 0.528) with only 1.76B tokens—a **96% reduction**. This shows that raw open-source data contains a weak training signal that can be concentrated via dedicated curation, and that scaling *usable* tokens matters more than simply increasing token budgets. The breakdown of our best open-source configuration is detailed in Table 4.
2. **Synthetic data provides better quality and a higher volume.** Because it is generated at
3. **Combining open-source and synthetic mixes may help at smaller scales.** Combined mixes can outperform synthetic-only mixes at smaller token budgets, but this advantage generally dematerialises as the budget is increased. For example, the best large combined mix uses more tokens and a lower threshold than the final synthetic mix (46.49B tokens, $\bar{z} \geq 0.41$ vs. 32.37B tokens, $\bar{z} \geq 0.68$), yet performs slightly worse. This suggests that once enough high-quality synthetic data is available, adding lower-quality open-source data can dilute the training signal.
4. **TopN capping is associated with better token efficiency.** Incorporating topN with threshold-only configurations as a means to limit the influ-

a larger scale, using a relatively high-quality teacher model, its unfiltered sentence pairs already benefit from higher quality scores than raw open-source pairs (Figure 3). This allows us to apply much stricter $\bar{z}$ score thresholds while still retaining sufficient training data. In competitive configurations, open-source mixes typically require relatively permissive thresholds around $\bar{z} \in [-0.5, 0.5]$, whereas synthetic mixes remain viable under substantially stricter thresholds, from $\bar{z} \geq 0.68$ to $\bar{z} \geq 1.2$. As a result, synthetic mixes dominate open-source mixes across nearly all training-token budgets.

ence of well-resourced language pairs tends to underperform in absolute performance although it tends to deliver a higher relative performance (per token). However, the effect is weak, therefore, restricted computational budgets should generally consider the topN capping option.

5. Bidirectional expansion as a robust strategy.

Across open-source, synthetic, and combined mixes, configurations with bidirectional expansion always improve compared to the equivalent configuration without expansion. This suggests that expanding selected examples to both translation directions improves coverage and keeps the data more balanced. Because filtering is still aligned with the final training direction, bidirectional expansion preserves QE reliability without the degradation caused by reversed filtering.

Based on these observations, we choose the best-performing synthetic mix (threshold + bidirectional expansion) as our final training dataset with 32.37B tokens and reaching **0.632 SSA-COMET**.

5.3 Which Dataset(s) Should We Use?

In this section, we evaluate several datasets in isolation to investigate their contribution to MT.

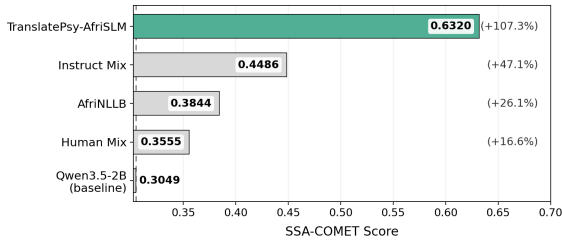


Figure 5: **Single-source SFT ablation on BOUQuET (Qwen3.5-2B)**. Each model is fine-tuned using a single data component and evaluated with SSA-COMET. The percentages indicate relative gains over the baseline.

AfriNLLB was the largest open-source, curated dataset¹³ to date hence we positioned it as our most competitive baseline in Figure 5. While it is more beneficial than the human translations, its contribution only has a moderate effect on African MT.

Human Mix is our highest-quality data, as it comes from projects with a high emphasis on human-quality translations. Figure 3 shows that the Human Mix receives higher QE scores than AfriNLLB, which helps explain the fact that it achieves comparable performance despite being

¹³Refer to Appendix A.5 for dataset preparation.

Model	Flores-200	BOUQuET	Smol
<i>General-purpose LLMs</i>			
Apertus-8B	0.3814	0.4024	0.3246
Qwen3.5-27B	0.5132	0.5336	0.4279
Qwen3.5-122B-A10B	0.5505	0.5716	0.4574
<i>Dedicated translation models</i>			
AfriNLLB-600M	0.5729	0.6038	0.4748
NLLB-1.3B	0.5878	0.6130	0.4850
NLLB-3.3B	0.5944	0.6178	0.4909
Hunyuan-MT-7B	0.4450	0.4451	0.3784
TranslateGemma-4B	0.3913	0.4069	0.3384
TranslateGemma-27B	0.5455	0.5677	0.4608
<i>African language-specialized models</i>			
AfriqueLlama-8B	0.5210	0.5620	0.4423
AfriqueGemma-12B	0.5531	0.5805	0.4608
AfriqueQwen-14B	0.5443	0.5718	0.4527
<i>Ours</i>			
TranslatePsy-AfriSLM-0.8B	0.5944	0.6223	0.4973
TranslatePsy-AfriSLM-2B	0.6070	0.6322	0.5074
TranslatePsy-AfriSLM-4B	0.6143	0.6391	0.5136

Table 3: **TranslatePsy-AfriSLM compared to LLMs and specialised translators** on Flores-200, BOUQuET, Smol (SSA-COMET). Full results in Table 17 and 18.

much smaller (60.2M versus 535M tokens). This suggests that human-quality data is highly valuable per token, but its limited scale makes it insufficient for effective SLM post-training on its own.

Instruct-Mix The primary reason for including this dataset in the final TranslatePsy-AfriSLM mix is to preserve the conversational skills of our SLMs, see Figure 13 for a short demo. However, we observe that it unexpectedly benefits machine translation as well, even more so than a dedicated AfriNLLB corpus (Figure 5). This is almost certainly due to the 2.3M examples (around 50% of total) of diverse tasks in African languages. This effect is similar to the AfriqueLLM (Yu et al., 2026) continued pre-training findings, which also observed an improvement in MT via instruction-following SFT.

TranslatePsy-AfriSLM Our data¹⁴ emerges as a powerful contributor to African MT. It combines scale and quality more effectively than prior sources: it retains enough data under strict filtering (Figure 4) and achieves higher QE scores than both AfriNLLB and the Human Mix (Figure 3). This makes it especially effective for African MT adaptation. Detailed statistics are given in Table 5.

5.4 Can SLMs Beat Frontier LLMs?

In this section, we combine the TranslatePsy-AfriSLM training dataset with the Instruct-Mix and

¹⁴The best synthetic mix, highlighted in Figure 4 (top right).

Asia-Europe Mix to improve real-world usability and contrast the performance of our SLMs with results from related methodologies.

Africa-IID Table 3 summarizes SSA-COMET performance across three benchmarks; full results across all four metrics are reported in Tables 17 and 18 (Appendix D). (1) **General-purpose LLMs** underperform, even at frontier scale. Qwen3.5-122B-A10B is surpassed by our 0.8B model on Flores-200, BOUQuET, and Smol, suggesting that model scale alone is insufficient for African MT. (2) **Dedicated translation models** also lag behind our SLMs. TranslatePsy-AfriSLM-0.8B outperforms AfriNLLB-600M, NLLB-3.3B and TranslateGemma-27B across all benchmarks. Notably, our 0.8B model matches NLLB-3.3B on Flores-200 and exceeds it on BOUQuET and Smol, despite using roughly a quarter of its parameters. This demonstrates that our carefully curated post-training data can produce powerful and efficient translation SLMs. (3) **African-centric models** narrow the gap but still do not match our 0.8B SLM. AfriqueGemma-12B and AfriqueQwen-14B remain below TranslatePsy-AfriSLM-0.8B on the reported benchmarks, while requiring substantially more compute at both training and inference time.

Statistical significance. Paired bootstrap tests confirm that the main rankings in Table 3 are statistically reliable (Appendix D.2). The results show that TranslatePsy-AfriSLM-0.8B significantly outperforms much larger LLM baselines on most settings, while TranslatePsy-AfriSLM-2B consistently surpasses dedicated NLLB models. We also observe statistically significant gains from TranslatePsy-AfriSLM-0.8B to 2B & 4B.

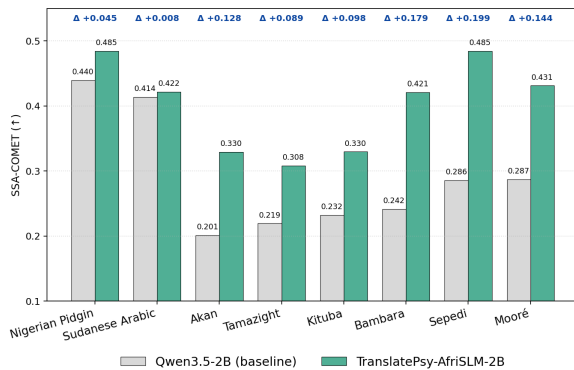


Figure 6: **Africa-OOD performance (SSA-COMET).** Qwen3.5-2B versus TranslatePsy-AfriSLM-2B. Scores are averaged over translation directions and all datasets.

Africa-OOD We use Africa-OOD to test whether training on our 19 target languages transfers to new languages. Under SSA-COMET, TranslatePsy-AfriSLM-2B improves over Qwen3.5-2B on all eight held-out languages, with particularly large gains for the lower-resource languages such as Sepedi, Bambara, and Akan (Figure 6). Further per-language analyses show similar trends across COMET-22, MetricX, and ChrF++ (Figure 11; Appendix D.1). These results suggest meaningful cross-lingual transfer beyond the training distribution, although less uniform than the IID gains.

Catastrophic Forgetting We include the Asia-Europe Mix (§3.4) to mitigate catastrophic forgetting on non-African languages. Figure 7 shows that SFT without this data causes substantial degradation on Asian and European languages, most notably under MetricX (−86.0%). Adding our Asia-Europe Mix reduces this to −10.3% and also mitigates degradation measured by COMET-22, SSA-COMET, and ChrF++. These results indicate that non-African parallel data helps preserve broader multilingual translation ability without sacrificing Africa-IID performance; in fact, we observe a small improvement on Africa-IID pairs (Appendix C).

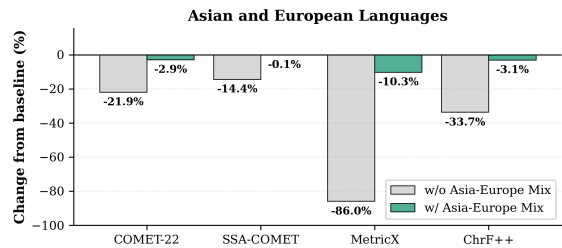


Figure 7: **Catastrophic forgetting on 38 Asian and European languages.** The bars show the percentage change from baseline after SFT, with/without the mix.

6 Conclusions

We have introduced **TranslatePsy-AfriSLM**, a comprehensive investigation of training data curation aimed at improving machine translation for 19 Sub-Saharan languages. Our results lead to two main conclusions: 1) Combining multiple QE metrics with robust $\bar{z}$ score normalization, and scoring examples in the same direction as training, yields more consistent filtering than relying on any single metric. This allows us to reduce the number of training tokens by up to 96% while maintaining comparable performance, 2) Across the 1B–50B token range, filtered synthetic data

dominates the quality-efficiency Pareto frontier, while open-source data appears to saturate early due to quality limitations. Given the current shortage of high-quality open-source parallel data for African MT, filtered synthetic generation seems to be the most practical path forward. Guided by these findings, we post-trained TranslatePsy-AfriSLM on our highest-quality data, outperforming much larger models, including TranslateGemma-27B and Qwen3.5-122B-A10B, with as few as 0.8B parameters. We open-source our data to support future work on making African MT a default capability in multilingual models.

Limitations

The Need for Human Evaluation. One notable finding that warrants future investigation is the challenge of determining absolute translation quality for African languages. Using the **reference-free quality estimators** from prior work, it appears that our TranslatePsy-AfriSLM data is comparable to human translated pairs in quality but it comes in quantities that are orders of magnitude larger. Still, the performance, as indicated by the **reference-based evaluation metrics** (produced by same model families as QE) suggests that we have not yet reached the translation quality of European and Asian languages. Therefore, conducting purely quantitative data scaling efforts using existing tools may run into **unknown absolute performance limitations**. The reason(s) behind this can only be resolved by analysing each step in the pipeline [data collection and training of QE models, correlation with human judgement, data collection and training of evaluation models, their correlation with ground truth], **leveraging expert human annotators** to identify any errors that have the capacity to compound over multiple steps during a large-scale data curation.

Ethics Statement

Data licensing All primary sources are open-source and publicly available under permissive attribution-based licenses — ODC-By v1.0 (Fine Translations, MADLAD-400) and CC BY 4.0 (MaLA Corpus), alongside research-permissive aggregates from OPUS and WMT22. The compiled TranslatePsy-AfriSLM datasets will be released under a similarly permissive license for safe, lawful, reproducible low-resource NLP research.

Dialectal representativeness Exact dialect tracking is unavailable for this web-curated corpus; models may over-reflect standardized written forms and under-represent regional dialects or oral traditions. We advise auditing trained models for localized sensitivity before deployment.

Synthetic content Because large internet repositories involve data recycling, mixing, duplication, and some LLM-based corrections and paraphrasing, even our "open-source" subset should be treated as at least partially synthetic (this is in addition to our purely synthetic translations).

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A Appendix

A.1 Synthetic and Open-Source Mixes

The corpora composing these mixes are described in §3.1. Here, we give more details of the processing steps and final data statistics. The preprocessing steps (up to decontamination) were implemented with NeMo Curator (NVIDIA, 2024). In Tables 4 and 5, "Clean" means before deduplication, "Preproc" after decontamination, "Filtered" after QE filtering, and "Bidir." after bidirectional expansion.

Processing steps:

1. **Document Splitting:**¹⁵ has divided documents into paragraphs (splitting at newlines).
2. **Cleaning:** removed Unicode artifacts, extra newlines, markup, and URLs.
3. **Language Identification:** via *AfroLID* (Adebara et al., 2022) for African languages and *FastText* (Joulin et al., 2016) for English.
4. **Sentence Splitting:**¹⁵ was performed using pySBD (Sadvilkar and Neumann, 2020).
5. **Filtering:** removed non-alphanumeric content and boilerplate text.
6. **Deduplication:** with exact & fuzzy matching.
7. **Eval-set Decontamination:** removed sentences similar to our test sets (§A.6).
8. **Synthetic Data Generation** The monolingual data (§ 3.1) was translated with the NLLB-3.3B model,¹⁶ decoded with the c-translate2 decoder¹⁷ with "int8_float16" quantisation and beam size 3. We discarded the 54B MoE model because the MoE architecture is not supported by fast decoders.
9. **Quality Estimation (QE):** For open-source data, the input to the QE stage is the parallel pre-processed text. For the synthetic data, it is the monolingual pre-processed source text together with its translation. For each pair, MetricX, AfriCOMET, SSA-COMET scores and the z-score were calculated in the source-target and target-source directions. Aligned filtering was applied (§3.2), and pairs with a z-score below the threshold were discarded.

¹⁵Document and sentence splitting was performed only on monolingual source texts to be translated for synthetic parallel data generation, because the teacher is not accurate/designed for paragraph-long input texts.

¹⁶<https://huggingface.co/facebook/nllb-200-3.3B>

¹⁷<https://github.com/OpenNMT/CTranslate2>

Lang.	Raw	Clean	Preproc	QE Filtered			Bidir.
				en→xx	xx→en	xx→yy	
afr	66,089	48,932	43,060	1,500	1,000	2,248	7,425
amh	41,109	36,935	30,542	1,020	776	2,841	6,518
hau	33,515	24,380	22,537	1,230	42	1,453	4,162
ibo	17,550	15,378	14,385	1,500	17	309	3,458
kin	21,855	16,425	15,484	1,500	0	280	2,989
lin	9,992	6,169	5,727	1,500	1	0	2,217
lug	14,257	8,175	7,528	1,302	4	80	1,922
mlg	11,106	8,336	7,829	1,394	10	19	2,477
nya	14,655	10,715	9,804	586	14	537	1,952
orm	11,949	5,703	5,517	892	8	7	1,175
sna	21,477	12,676	11,744	1,391	11	510	3,488
som	26,157	22,432	20,068	1,388	39	292	3,425
sot	4,517	3,676	3,491	23	1	0	55
swa	37,664	30,784	28,223	1,500	41	697	5,947
tsn	14,595	9,325	8,520	1,609	25	63	3,085
wol	4,342	3,506	3,280	868	1	14	1,218
xho	22,971	14,435	13,534	1,500	13	200	3,595
yor	13,909	9,052	8,091	411	67	69	1,999
zul	24,476	13,709	12,950	1,132	15	3	3,196
others	15,004	14,828	14,292	1,500	1,000	0	5,000
Total	427,187	315,571	286,606	23,747	3,085	9,623	65,304

Table 4: **Open-Source mix statistics** by language (line counts in thousands).

A.2 Human Mix

We compiled 352,582 human-translated, high-quality parallel sentences focused on African languages from AfriDOC-MT (Alabi et al., 2025) and SMOL¹⁸ (Caswell et al., 2025). Table 8 summarizes the high-level statistics. The SmolDoc subset of SMOL¹⁹ was flattened to sentence-level pairs.

Processing Steps:

1. **Bidirectional expansion:** Both directions (en→xx and xx→en) were used for training.
2. **Global exact deduplication:** Identical pairs were removed.
3. **Eval-set decontamination:** Removed sentences similar to our evaluation data.
4. **Global approximate deduplication:** Min-Hash applied to remove similar examples.

Filter Type	Kept	Removed	Retained %
Raw sentence pairs	180,540	–	100.00%
(1) After bidirectional expansion	361,080	–	200.00%
(2) After global exact deduplication	357,584	3,496	99.03%
(3) After eval-set decontamination	357,402	182	99.95%
(4) After global approximate deduplication	352,582	4,820	98.65%

Table 8: Filtering summary for the human translations.

A.3 Instruct Mix

Afri-Instruct Mix This dataset is a heterogeneous mixture of open-ended instruction data aggregated from 11 publicly available HuggingFace datasets (Singh et al., 2024; Taori et al., 2023; Devine et al., 2026; Owusu, 2025; Muhammad et al., 2023; Adelani et al., 2023; Ogundepo et al.,

¹⁸Excluding our evaluation set (**smolsent**).

¹⁹<https://huggingface.co/datasets/google/smol>

Lang.	Monolingual			Parallel, QE Filtered		
	Raw	Clean	Preproc	eng→xx	xx→eng	Bidir.
eng	145,726	145,688	136,482	—	—	—
afr	24,062	23,384	11,139	1,009	1,875	6,974
amh	3,704	3,576	3,379	4,897	569	10,786
hau	5,002	4,729	4,420	6,697	1,306	17,261
ibo	2,450	2,351	2,222	5,964	639	12,659
kin	5,211	4,699	4,219	2,536	847	8,102
lin	153	145	135	5,056	36	7,482
lug	378	358	332	5,586	48	10,089
mlg	3,521	2,690	2,464	1,010	470	4,168
nya	2,292	1,846	1,768	3,575	439	8,060
orm	75	70	68	4,817	10	7,580
sna	426	344	335	2,490	92	5,829
som	6,981	6,616	5,967	2,744	851	7,381
sot	1,980	1,851	1,754	4,576	503	8,495
swa	18,080	17,559	15,834	3,598	2,776	17,670
tsn	22	21	20	3,506	7	6,033
wol	45	43	39	13,256	7	23,615
xho	2,219	2,040	1,915	4,414	465	7,962
yor	2,345	2,109	2,007	13,464	1,128	27,140
zul	2,118	1,737	1,662	9,115	519	18,367
Total	226,790	221,856	196,162	98,312	12,589	215,653

Table 5: **Synthetic-Mix stats** (TranslatePsy-AfriSLM training data) by language (line counts in thousands).

Quality estimator	C22	SSA	MX	ChrF++
eng-xx (<i>English</i> → <i>African language(s)</i>)				
z-score	0.737	0.617	4.52	45.7
SSA-COMET	0.736	0.620	4.73	45.9
AfriCOMET	0.735	0.609	4.87	45.3
MetricX	0.730	0.605	4.29	45.2
xx-eng (<i>African language(s)</i> → <i>English</i>)				
z-score	0.754	0.581	5.30	52.0
SSA-COMET	0.746	0.581	5.79	51.9
AfriCOMET	0.756	0.579	5.40	52.3
MetricX	0.749	0.573	5.27	51.7

Table 6: **Translation quality with individual versus unified QE metrics** over Flores-200 test set.

2023; Adelani et al., 2025; Ojo et al., 2025; Hasan et al., 2021; Ding et al., 2023). Table 9 summarizes each data source. The final mixture contains 2,306,800 examples designed to maintain/extend essential instruction-following abilities to African MT, e.g. open-ended instruction and chat data, which contributes 1,553,944 examples (67.36%), followed by code-assistance data with 521,389 examples (22.60%). Smaller portions come from classification and other structured prediction tasks with 123,448 examples (5.35%), summarization and headline generation with 74,581 examples (3.23%), and question answering plus cross-lingual QA with 33,438 examples (1.45%).

Processing Steps:

1. **Format standardization:** Each dataset was

Quality estimator	C22	SSA	MX	ChrF++
eng-xx (<i>English</i> → <i>African language(s)</i>)				
z-score	0.655	0.503	8.88	28.0
SSA-COMET	0.652	0.505	9.22	28.0
AfriCOMET	0.653	0.495	9.26	27.8
MetricX	0.650	0.491	8.58	27.9
xx-eng (<i>African language(s)</i> → <i>English</i>)				
z-score	0.570	0.500	10.25	29.1
SSA-COMET	0.565	0.498	10.65	29.2
AfriCOMET	0.570	0.497	10.40	29.3
MetricX	0.568	0.491	10.14	29.2

Table 7: **Translation quality with individual versus unified QE metrics** over Smol test set.

converted to a multi-turn chat format with system, user, and assistant roles. Appropriate system prompts were added (e.g., “You are a sentiment classifier” for AfriSenti, “You are a summarization assistant” for XLSum), etc.

2. **Quality filtering:** Samples with empty user or assistant messages were removed.
3. **Task expansion:** AfriQA was expanded into three task types: machine translation pairs, monolingual QA, and cross-lingual QA, tripling its effective sample count.
4. **Sampling:** For Afri-Code datasets, a maximum of 30,000 samples per language were randomly sampled to reduce repetition.

Dataset	Source	Count	Languages
African-UltraChat	masakhane/african-ultrachat	54,994	11
African-Alpaca	masakhane/african-translated-alpaca	832,029	16
Kakugo	ptrdvn/kakugo-{lang}	464,569	12
Aya Dataset	CoherentLabs/aya_dataset	202,352	65
Afri-Code	michsethowsu/Code-170k-{lang}	521,389	18
AfriSenti	shmuhammad/AfriSenti-twitter-sentiment	83,688	8
MasakhaNEWS	masakhane/masakhanews	21,296	12
XLSum	csebutnlp/xlsun	53,285	7
AfriADR	masakhane/AfriADR	26,110	3
AfriXNLI	masakhane/afrixnli	13,650	13
AfriQA	masakhane/afriqa	33,438	7
Total	-	2,306,800	-

Table 9: Data sources for the Afri-Instruct data mix.

General-Instruct Mix Additional instruction-following training examples (mostly English-centric) come from two public HuggingFace datasets: **smoltalk2**, the post-training data of SmolLM3 (Bakouch et al., 2025) and **Dolci-Instruct**, the OLMO-3 supervised fine-tuning data (Olmo et al., 2025). Table 10 summarizes each data source. Exact deduplication removed 85,982 examples. Due to the exclusion of reasoning data in TranslatePsy-AfriSLM training, it does not retain the ‘thinking’ capability of its base model.

Processing Steps:

1. **Format standardization:** Datasets were converted to a unified chat format. A system prompt (“You are a helpful assistant.”) was added to conversations from Dolci-Instruct.
2. **Exact deduplication:** Identical examples were removed.

Dataset	Source	Splits
SmolTalk2	HuggingFaceTB/smol1talk2 (SFT)	multilingual_8languages_lang_5_no_think
		smollm3_systemchats_30k_no_think
		smollm3_everyday_conversations_no_think
		smollm3_explore_instruct_rewriting_no_think
		smollm3_smol_rewrite_no_think
Dolci-Instruct	allenai/Dolci-Instruct-SFT-No-Tools	train
Total	-	2,308,569

Table 10: Data sources for the General Instruct mix.

A.4 Asia-Europe Mix

In order to mitigate catastrophic forgetting, we included a broad selection of 38 medium-high resource languages²⁰ from OPUS-100 (Zhang et al., 2020; Tiedemann, 2012). The final dataset comprises 24,114,303 parallel sentences. We only use en → xx pairs as the xx → en direction is robust to large-scale African post-training. Including this data has negligible impact on African translation quality, while it plays a crucial role in preserving Asian and European performance (Appendix C). Table 11 provides an aggregate summary while Table 12 in summarizes the per-language statistics.

Processing Steps:

1. **Quality filtering:** Samples were filtered to only those with a mean bidirectional COMET²¹ (Rei et al., 2022b) score ≥ 0.6 .
2. **Approximate deduplication:** MinHash was applied using 128 permutations and a Jaccard similarity of 0.8 on character 4-grams.
3. **Eval-set decontamination:** Sentences similar to our evaluation sets were removed.
4. **Global approximate deduplication:** BPE-unigram MinHash deduplication (256 permutations, Jaccard threshold 0.8) was applied globally to remove similar examples.

Filter Type	Kept	Removed	Retained %
Raw sentence pairs	36,347,268	-	100.00%
After COMET mean ≥ 0.6	30,112,876	6,234,392	82.85%
After pair-level (local) MinHash deduplication	25,439,466	4,673,410	84.48%
After eval-set decontamination	25,429,129	10,337	99.96%
After MinHash deduplication	24,114,303	1,314,826	94.83%

Table 11: Filtering summary for Europe/Asia mix.

Pair	RAW	COMET	DEDUP	Pair	RAW	COMET	DEDUP
en-zh	1,000k	837k	767k	en-sv	1,000k	828k	725k
en-ja	1,000k	788k	669k	en-hr	1,000k	829k	738k
en-it	1,000k	843k	752k	en-is	1,000k	650k	485k
en-ru	1,000k	817k	751k	en-lt	1,000k	879k	650k
en-es	1,000k	879k	791k	en-ms	1,000k	815k	634k
en-tr	1,000k	880k	778k	en-id	1,000k	860k	720k
en-fr	1,000k	839k	789k	en-si	979k	829k	475k
en-pl	1,000k	787k	695k	en-hi	534k	474k	298k
en-ar	1,000k	809k	751k	en-bn	1,000k	896k	604k
en-uk	1,000k	803k	578k	en-ur	754k	727k	603k
en-pt	1,000k	848k	748k	en-fa	1,000k	813k	730k
en-sk	1,000k	860k	716k	en-kk	80k	57k	41k
en-de	1,000k	773k	720k	en-ro	1,000k	831k	725k
en-hu	1,000k	828k	732k	en-bg	1,000k	814k	718k
en-el	1,000k	822k	735k	en-cs	1,000k	814k	720k
en-ko	1,000k	726k	617k	en-da	1,000k	820k	712k
en-vi	1,000k	863k	729k	en-lv	1,000k	879k	652k
en-th	1,000k	841k	721k	en-nl	1,000k	830k	747k
en-fi	1,000k	810k	732k	en-et	1,000k	814k	691k

Table 12: Language statistics for Asia-Europe data.

A.5 AfriNLLB

We apply only test data decontamination and minimal approximate deduplication, see Table 13.

Processing Steps:

1. **Approximate deduplication:** MinHash was applied using 128 permutations and a Jaccard similarity of 0.8 on character 4-grams.
2. **Eval-set decontamination:** Sentences similar to our evaluation sets were removed.

Filter Type	Kept	Removed	Retained %
Original AfriNLLB pairs	3,218,822	-	100.00%
After eval-set decontamination	3,206,918	11,904	99.63%
After approximate deduplication	3,129,175	77,743	97.58%

Table 13: AfriNLLB preprocessing steps.

A.6 Data Decontamination

As an essential part of our preprocessing steps, we decontaminate our training data against all evaluation sentences, languages and dataset splits, totaling **over 850K sentences**. We used a BPE-unigram (Qwen3.5) MinHash (LSH) deduplication with a Jaccard similarity threshold of 0.9 to filter training data on an individual sentence level rather than a pair level for the most granular detection possible.

B Additional Experimental Setup

B.1 Translation Prompt

Training and evaluation examples were structured as shown in Figure 8 where SOURCE_LANG and TARGET_LANG denote the source and target language **names** while SOURCE_TEXT and TARGET_TEXT represent the input/output **texts** in each language, respectively. Finally, a model-specific chat template was applied.

²⁰<https://huggingface.co/datasets/Helsinki-NLP/opus-100>

²¹<https://huggingface.co/Unbabel/wmt22-cometkiwi-da>

MT Prompt Template

System Prompt:

You are a professional SOURCE_LANG to TARGET_LANG translator. Your goal is to accurately convey the meaning and nuances of the original SOURCE_LANG text while adhering to TARGET_LANG grammar, vocabulary, and cultural sensitivities. Produce only the TARGET_LANG translation, without any additional explanations or commentary.

User:

Please translate the following SOURCE_LANG text into TARGET_LANG: SOURCE_TEXT. Translation:

Assistant:

TARGET_TEXT

Figure 8: Training and evaluation prompt template.

B.2 Training Hyperparameters

Optimization. Full fine-tuning uses peak learning rate of 1.25×10^{-5} ; fused AdamW, a linear schedule with 1% warmup, gradient clipping at 1.0, and gradient checkpointing, with a global batch size of 256.

Data formatting. All training samples are first converted into the translation prompt format shown in Figure 8, and then wrapped with the model-specific chat template. Sequences exceeding 2,048 tokens are filtered, and remaining samples are packed via best-fit decreasing.

Infrastructure. All experiments use PyTorch, HuggingFace Transformers, TRL, and DeepSpeed ZeRO-2 in bfloat16 on 32 NVIDIA H100 GPUs.

C Catastrophic Forgetting Analysis

We further analyze whether the Asia-Europe Mix mitigates catastrophic forgetting on non-African languages without compromising African MT performance. The mix contains bidirectional parallel data between English and non-African languages, covering both Asian²² and European²³ language groups.

Figure 9 shows that fine-tuning without the Asia-Europe Mix substantially degrades performance on Asian and European language pairs, indicating catastrophic forgetting outside the African training distribution. Adding the mix consistently reduces this degradation across all metrics and both language groups. For example, the MetricX-24 drop

²²Asian: sin, hin, ben, urd, fas, kaz, tur, zho, jpn, kor, vie, tha, ind, msa, fil.

²³European: fra, spa, ita, por, ron, ell, lit, lav, rus, ukr, pol, slk, hrv, ces, srp, bul, deu, swe, isl, dan, nld, hun, fin, est.

AFRICA-OD lang.	Flores-200	BOUQuET	Smol
Nigerian Pidgin (pcm)		✓	✓
Sudanese Arabic (apd)			✓
Akan (aka)	✓	✓	✓
Tamazight (ber)	✓	✓	✓
Kituba (ktu)			✓
Bambara (bam)	✓	✓	✓
Sepedi (nso)	✓	✓	✓
Mooré (mos)	✓	✓	✓
# Langs.	5	6	8
# Directions	10	12	16

Table 14: AFRICA-OD language coverage by evaluation set. The number of directions counts both eng-to-xx and xx-to-eng directions.

is reduced from -95.1% to -16.8% for Asian languages and from -81.0% to -6.7% for European languages. This retention does not come at the cost of Africa-IID performance: models with and without the Asia-Europe Mix achieve nearly identical gains on African language pairs across all metrics. We therefore retain the Asia-Europe Mix in the final TranslatePsy-AfriSLM recipe, not to improve African MT directly, but to preserve broader multilingual translation while maintaining the gains on the Africa-IID group.

D Additional Results

D.1 Per-Language Performance Analysis

Africa-IID Figure 10 shows per-language results on the Africa-IID BOUQuET benchmark, averaging scores over both eng-xx and xx-eng directions. TranslatePsy-AfriSLM-2B consistently improves over the Qwen3.5-2B baseline across the 19 in-distribution languages and all four evaluation metrics. The largest gains appear for low-baseline languages such as Oromo, Malagasy, Lingala, Tswana, and Zulu, while Afrikaans shows smaller gains due to its stronger baseline.

Africa-OD Figure 11 reports results for Africa-OD languages held out from fine-tuning. Because Flores-200, BOUQuET, and Smol cover different subsets of OOD languages, scores are averaged over both directions and available evaluation sets, as summarized in Table 14. Although the OOD languages are held out from fine-tuning, TranslatePsy-AfriSLM-2B still improves overall performance, suggesting meaningful transfer beyond the languages seen during post-training. The strongest gains appear for Sepedi, Bambara, Akan, and Mooré, while improvements are smaller and more heterogeneous than in the IID setting, with

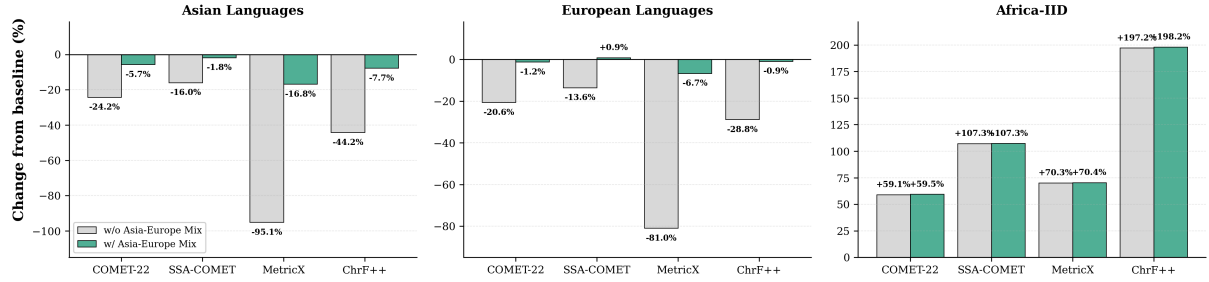


Figure 9: **Effect of the Asia-Europe Mix on catastrophic forgetting and Africa-IID performance.** Bars show the percentage change from the Qwen3.5-2B baseline after SFT. The Asia-Europe Mix substantially reduces degradation on Asian and European language pairs while preserving nearly identical gains on the target Africa-IID pairs.

some metric-specific regressions for Nigerian Pidgin, Sudanese Arabic, and Tamazight.

D.2 Statistical Significance (Bootstrap)

We assess statistical significance using paired bootstrap resampling over sentence pairs. For each model pair, dataset, and metric, we align common translation directions, pair sentence-level scores, and draw $B = 10,000$ bootstrap samples with replacement. We report Δ as the mean score difference and compute p as the fraction of bootstrap samples that do not support the observed direction of improvement. For MetricX, lower scores are better; for all other metrics, higher scores are better.

Table 16 shows that the main rankings are highly stable across benchmarks and metrics. TranslatePsy-AfriSLM-0.8B substantially outperforms the much larger general-purpose Qwen3.5-122B on SSA-COMET, MetricX, and ChrF++, and remains competitive on COMET-22. It also consistently surpasses TranslateGemma-27B and AfriGemma-12B across all reported settings. Against NLLB, TranslatePsy-AfriSLM-0.8B is competitive but mixed, whereas TranslatePsy-AfriSLM-2B significantly outperforms both NLLB-1.3B and NLLB-3.3B across every benchmark-metric combination. Within the TranslatePsy-AfriSLM family, scaling from 0.8B to 2B and from 2B to 4B yields monotonic and statistically significant gains. Overall, the bootstrap results confirm that the observed improvements are stable under paired sentence-level resampling rather than being driven by evaluation noise.

Data Source	Configuration	Threshold	Tokens	Threshold	Tokens	Threshold	Tokens
Open-Source	threshold	-0.50	6.57B	0.00	3.79B	0.50	1.76B
	threshold + top- N	-0.28	5.28B	0.16	2.88B	0.61	1.20B
	threshold + bidirectional	-0.50	14.12B	0.00	7.86B	0.50	3.40B
	threshold + top- N + bidirectional	-0.28	9.95B	0.16	5.15B	0.61	2.03B
Synthetic	threshold	0.68	16.75B	0.93	7.28B	1.19	2.80B
	threshold + top- N	0.71	12.43B	0.93	4.49B	1.18	1.44B
	threshold + bidirectional	0.68	32.37B	0.93	15.05B	1.19	6.16B
	threshold + top- N + bidirectional	0.71	23.62B	0.93	9.09B	1.18	3.32B
Open-Source + Synthetic	threshold	0.41	23.32B	0.67	11.08B	0.97	4.56B
	threshold + top- N	0.55	17.71B	0.78	7.37B	1.07	2.64B
	threshold + bidirectional	0.41	46.49B	0.67	22.92B	0.97	9.56B
	threshold + top- N + bidirectional	0.55	33.57B	0.78	14.24B	1.07	5.35B

Table 15: **Threshold values and training token counts for each data source, configuration, and filtering level** (for the datasets plotted in Figure 4). Since the threshold may vary depending on the data mix and language direction, and topN changes the effective threshold, the value shown is an average weighted by the number of example pairs at each actual threshold. The bolded configuration indicates the TranslatePsy-AfriSLM data mix used for final training.



Figure 10: **AFRICA-IID BOUQuET results.** Per-language scores for 19 target African languages on BOUQuET, comparing Qwen3.5-2B and TranslatePsy-AfriSLM-2B across four evaluation metrics. Scores are averaged over both translation directions where available. Δ = TranslatePsy-AfriSLM-2B – baseline model scores.

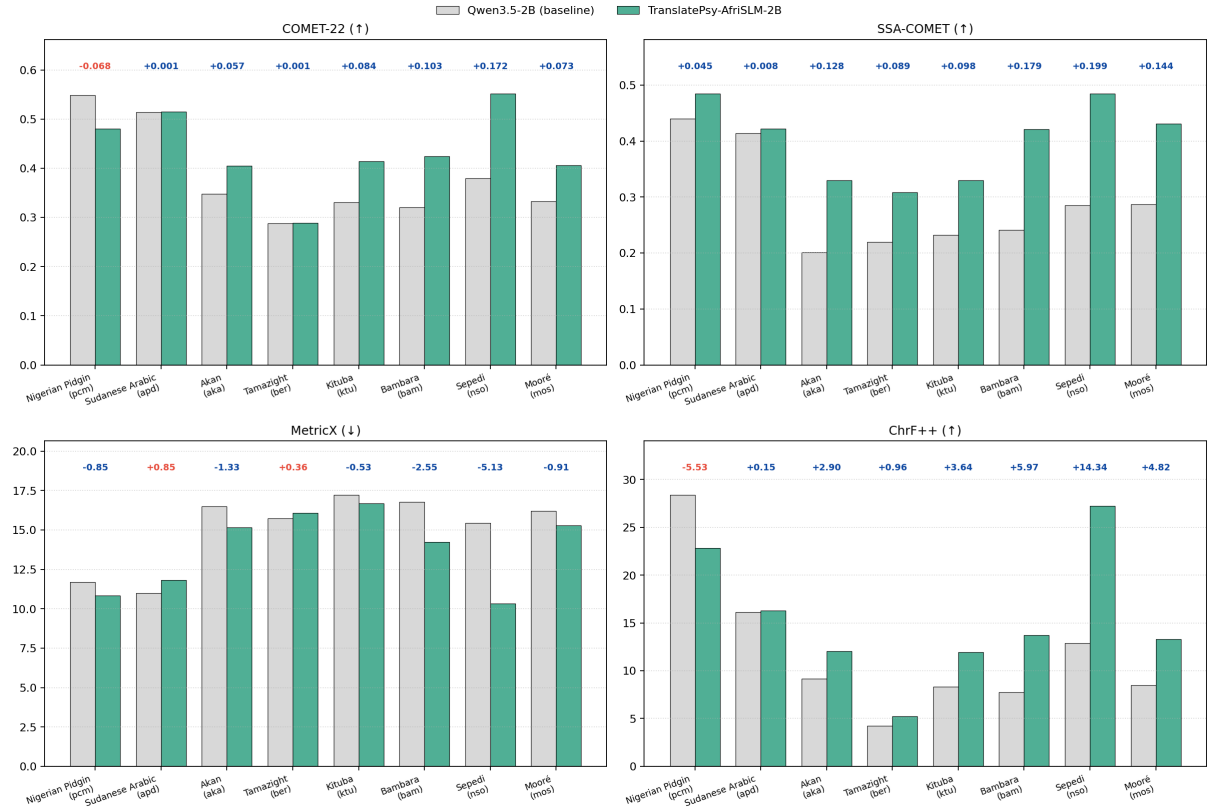


Figure 11: **AFRICA-ODD results across all evaluation metrics.** Per-language scores for held-out African languages, comparing Qwen3.5-2B and TranslatePsy-AfriSLM-2B. Scores are averaged over both translation directions and over the available evaluation sets among Flores-200, BOUQuET, and Smol; language coverage differs by evaluation set as shown in Table 14. Δ = TranslatePsy-AfriSLM-2B – baseline model scores.

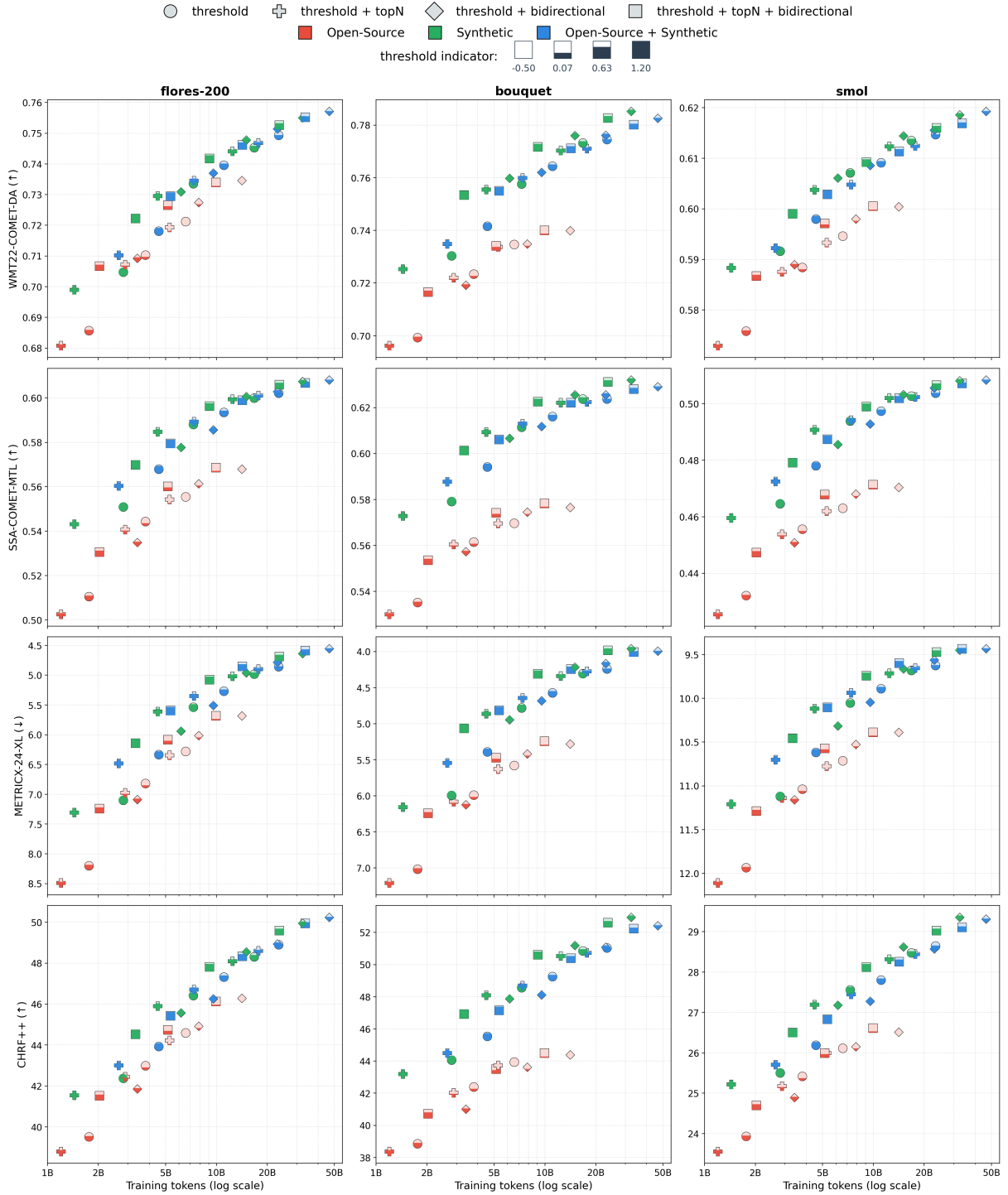


Figure 12: **Scaling runs as QE quality and training budgets increase**, evaluated on Flores-200, BOUQuET, and Smol (columns). Each row reports a different metric, each point corresponds to a model trained with a particular data source and filtering configuration. Colours distinguish filtered, synthetic, and filtered+synthetic data, while marker shapes indicate the filtering setup. MetricX is lower-is-better; all other metrics are higher-is-better. Filtering thresholds are represented by an increasing fill, i.e. the fuller the shape, the higher the threshold.

Comparison	Metric	FLORES-200		BOUQuET		SMOL	
		Δ	p	Δ	p	Δ	p
<i>vs. General-purpose LLM</i>							
TranslatePsy-AfriSLM-0.8B vs. Qwen3.5-122B	COMET-22	-0.0001	0.452	+0.0135	<0.001	+0.0017	<0.001
TranslatePsy-AfriSLM-0.8B vs. Qwen3.5-122B	SSA-COMET	+0.0413	<0.001	+0.0379	<0.001	+0.0286	<0.001
TranslatePsy-AfriSLM-0.8B vs. Qwen3.5-122B	MetricX	-0.548	<0.001	-0.723	<0.001	-0.506	<0.001
TranslatePsy-AfriSLM-0.8B vs. Qwen3.5-122B	ChrF++	+3.30	<0.001	+5.33	<0.001	+1.29	<0.001
<i>vs. Dedicated Translation</i>							
TranslatePsy-AfriSLM-0.8B vs. NLLB-1.3B	COMET-22	-0.0039	<0.001	+0.0011	0.017	+0.0021	<0.001
TranslatePsy-AfriSLM-0.8B vs. NLLB-1.3B	SSA-COMET	+0.0066	<0.001	+0.0086	<0.001	+0.0122	<0.001
TranslatePsy-AfriSLM-0.8B vs. NLLB-1.3B	MetricX	+0.038	0.002	-0.154	<0.001	-0.242	<0.001
TranslatePsy-AfriSLM-0.8B vs. NLLB-1.3B	ChrF++	-0.40	<0.001	+0.48	<0.001	-0.01	0.395
TranslatePsy-AfriSLM-0.8B vs. NLLB-3.3B	COMET-22	-0.0093	<0.001	+0.0064	<0.001	+0.0011	0.002
TranslatePsy-AfriSLM-0.8B vs. NLLB-3.3B	SSA-COMET	+0.0000	0.483	+0.0133	<0.001	+0.0124	<0.001
TranslatePsy-AfriSLM-0.8B vs. NLLB-3.3B	MetricX	+0.284	<0.001	-0.007	0.314	+0.004	0.385
TranslatePsy-AfriSLM-0.8B vs. NLLB-3.3B	ChrF++	-1.45	<0.001	+0.12	0.101	-0.19	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-1.3B	COMET-22	+0.0081	<0.001	+0.0119	<0.001	+0.0091	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-1.3B	SSA-COMET	+0.0193	<0.001	+0.0189	<0.001	+0.0227	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-1.3B	MetricX	-0.522	<0.001	-0.565	<0.001	-0.624	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-1.3B	ChrF++	+1.28	<0.001	+2.42	<0.001	+0.75	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-3.3B	COMET-22	+0.0027	<0.001	+0.0172	<0.001	+0.0081	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-3.3B	SSA-COMET	+0.0126	<0.001	+0.0237	<0.001	+0.0229	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-3.3B	MetricX	-0.276	<0.001	-0.419	<0.001	-0.378	<0.001
TranslatePsy-AfriSLM-2B vs. NLLB-3.3B	ChrF++	+0.24	<0.001	+2.06	<0.001	+0.57	<0.001
TranslatePsy-AfriSLM-0.8B vs. TGemma-27B	COMET-22	+0.0097	<0.001	+0.0272	<0.001	+0.0018	<0.001
TranslatePsy-AfriSLM-0.8B vs. TGemma-27B	SSA-COMET	+0.0489	<0.001	+0.0547	<0.001	+0.0365	<0.001
TranslatePsy-AfriSLM-0.8B vs. TGemma-27B	MetricX	-0.945	<0.001	-1.190	<0.001	-0.643	<0.001
TranslatePsy-AfriSLM-0.8B vs. TGemma-27B	ChrF++	+5.08	<0.001	+7.79	<0.001	+2.33	<0.001
<i>vs. African-Specialized</i>							
TranslatePsy-AfriSLM-0.8B vs. AfriGemma-12B	COMET-22	+0.0301	<0.001	+0.0204	<0.001	+0.0212	<0.001
TranslatePsy-AfriSLM-0.8B vs. AfriGemma-12B	SSA-COMET	+0.0414	<0.001	+0.0306	<0.001	+0.0284	<0.001
TranslatePsy-AfriSLM-0.8B vs. AfriGemma-12B	MetricX	-0.664	<0.001	-0.531	<0.001	-0.444	<0.001
TranslatePsy-AfriSLM-0.8B vs. AfriGemma-12B	ChrF++	+3.57	<0.001	+3.19	<0.001	+2.03	<0.001
<i>Scaling Effect</i>							
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-2B	COMET-22	+0.0077	<0.001	+0.0080	<0.001	+0.0059	<0.001
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-2B	SSA-COMET	+0.0073	<0.001	+0.0071	<0.001	+0.0063	<0.001
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-2B	MetricX	-0.320	<0.001	-0.287	<0.001	-0.260	<0.001
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-2B	ChrF++	+1.14	<0.001	+1.33	<0.001	+0.65	<0.001
TranslatePsy-AfriSLM-2B vs. TranslatePsy-AfriSLM-0.8B	COMET-22	+0.0120	<0.001	+0.0116	<0.001	+0.0085	<0.001
TranslatePsy-AfriSLM-2B vs. TranslatePsy-AfriSLM-0.8B	SSA-COMET	+0.0126	<0.001	+0.0163	<0.001	+0.0152	<0.001
TranslatePsy-AfriSLM-2B vs. TranslatePsy-AfriSLM-0.8B	MetricX	-0.560	<0.001	-0.552	<0.001	-0.441	<0.001
TranslatePsy-AfriSLM-2B vs. TranslatePsy-AfriSLM-0.8B	ChrF++	+1.68	<0.001	+1.96	<0.001	+0.83	<0.001
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-0.8B	COMET-22	+0.0197	<0.001	+0.0196	<0.001	+0.0143	<0.001
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-0.8B	SSA-COMET	+0.0199	<0.001	+0.0234	<0.001	+0.0214	<0.001
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-0.8B	MetricX	-0.880	<0.001	-0.840	<0.001	-0.701	<0.001
TranslatePsy-AfriSLM-4B vs. TranslatePsy-AfriSLM-0.8B	ChrF++	+2.82	<0.001	+3.29	<0.001	+1.49	<0.001

Table 16: **Paired bootstrap significance tests** ($B = 10,000$). Δ = score difference (ours – baseline). Baseline comparisons use 0.8B models. Blue Δ = ours better; red = ours worse (MetricX: lower is better, negative Δ is blue).

Model	Flores-200		BOUQuET		Smol	
	SSA (↑)	MX (↓)	SSA (↑)	MX (↓)	SSA (↑)	MX (↓)
<i>General-purpose LLMs/SLMs</i>						
Apertus-8B	0.3814	11.471	0.4024	10.385	0.3246	14.233
Qwen3-4B	0.2426	15.427	0.2565	14.772	0.2117	17.273
Qwen3-8B	0.2655	15.327	0.2760	14.725	0.2275	17.336
Qwen3.5-0.8B	0.2967	15.974	0.3320	14.902	0.2454	17.595
Qwen3.5-2B	0.2885	14.344	0.3049	13.317	0.2498	15.926
Qwen3.5-4B	0.3863	11.416	0.3984	10.567	0.3301	13.868
Qwen3.5-9B	0.4523	9.758	0.4664	8.927	0.3811	13.216
Qwen3.5-27B	0.5132	7.250	0.5336	6.468	0.4279	11.420
Qwen3.5-122B-A10B	0.5505	5.920	0.5716	5.318	0.4574	10.515
<i>Dedicated translation models</i>						
AfriNLLB-600M	0.5729	5.916	0.6038	4.831	0.4748	10.476
NLLB-600M	0.5718	5.752	0.6019	4.803	0.4723	10.430
NLLB-1.3B	0.5878	5.159	0.6130	4.448	0.4850	10.036
NLLB-3.3B	0.5944	4.913	0.6178	4.264	0.4909	9.841
Hunyuan-MT-7B	0.4450	10.779	0.4451	10.659	0.3784	13.949
TranslateGemma-4B	0.3913	11.086	0.4069	10.219	0.3384	14.000
TranslateGemma-12B	0.5037	7.718	0.5228	6.926	0.4299	11.501
TranslateGemma-27B	0.5455	6.142	0.5677	5.506	0.4608	10.468
<i>African language-specialized models</i>						
AfriqueLlama-8B	0.5210	7.057	0.5620	5.679	0.4423	10.974
AfriqueGemma-4B	0.5276	6.692	0.5587	5.716	0.4410	10.938
AfriqueGemma-12B	0.5531	5.862	0.5805	5.093	0.4608	10.397
AfriqueQwen-8B	0.5222	6.871	0.5549	5.748	0.4394	10.959
AfriqueQwen-14B	0.5443	6.243	0.5718	5.408	0.4527	10.748
<i>Ours</i>						
TranslatePsy-AfriSLM-0.8B	0.5944	5.197	0.6223	4.316	0.4973	9.823
TranslatePsy-AfriSLM-2B	0.6070	4.637	0.6322	3.940	0.5074	9.458
TranslatePsy-AfriSLM-4B	0.6143	4.317	0.6391	3.701	0.5136	9.232

Table 17: **SSA-COMET** and **MetricX** results on Flores-200, BOUQuET, and Smol.

Model	Flores-200		BOUQuET		Smol	
	C22 (↑)	ChrF++ (↑)	C22 (↑)	ChrF++ (↑)	C22 (↑)	ChrF++ (↑)
<i>General-purpose LLMs</i>						
Apertus-8B	0.5883	30.45	0.6025	27.84	0.4912	18.02
Qwen3-4B	0.4046	17.34	0.4305	14.25	0.3607	10.74
Qwen3-8B	0.4566	20.64	0.4792	17.29	0.4002	12.50
Qwen3.5-0.8B	0.3656	14.02	0.3920	11.17	0.3255	8.60
Qwen3.5-2B	0.4724	20.95	0.4934	17.82	0.4143	12.90
Qwen3.5-4B	0.6021	30.58	0.6122	27.87	0.5103	18.59
Qwen3.5-9B	0.6688	36.73	0.6801	34.96	0.5566	22.19
Qwen3.5-27B	0.7144	41.76	0.7264	41.22	0.5859	24.93
Qwen3.5-122B-A10B	0.7373	44.42	0.7494	44.32	0.6045	26.68
<i>Dedicated translation models</i>						
AfriNLLB-600M	0.7315	47.67	0.7671	49.81	0.6008	28.38
NLLB-600M	0.7340	46.81	0.7654	49.03	0.6028	27.68
NLLB-1.3B	0.7471	48.72	0.7753	50.77	0.6101	28.52
NLLB-3.3B	0.7525	49.79	0.7795	52.36	0.6139	29.03
Hunyuan-MT-7B	0.5056	23.49	0.5211	19.55	0.4391	15.48
TranslateGemma-4B	0.5487	24.78	0.5699	23.12	0.4644	15.21
TranslateGemma-12B	0.7125	39.48	0.7211	38.92	0.5983	24.55
TranslateGemma-27B	0.7335	43.32	0.7499	43.70	0.6102	26.24
<i>African language-specialized models</i>						
AfriqueLlama-8B	0.6689	37.59	0.7179	41.58	0.5630	23.03
AfriqueGemma-4B	0.6916	40.74	0.7223	42.49	0.5688	23.60
AfriqueGemma-12B	0.7131	44.52	0.7429	46.26	0.5847	25.94
AfriqueQwen-8B	0.6754	38.23	0.7168	41.08	0.5659	23.06
AfriqueQwen-14B	0.6955	41.58	0.7297	43.69	0.5754	24.64
<i>Ours</i>						
TranslatePsy-AfriSLM-0.8B	0.7432	48.30	0.7770	51.30	0.6120	28.54
TranslatePsy-AfriSLM-2B	0.7552	50.00	0.7870	53.13	0.6187	29.32
TranslatePsy-AfriSLM-4B	0.7629	51.13	0.7941	54.33	0.6237	29.97

Table 18: COMET-22 and ChrF++ results on Flores-200, BOUQuET and Smol.

Multilingual MT Example

User: hello, I need to translate this “Esika nini station ya engbunduka ezali?” into English.

Assistant: “Where is the train station?” is the English translation of “Esika nini station ya engbunduka ezali?”

User: Can you translate the same sentence into Swahili, German and Amharic?

Assistant: “Where is the train station?” in Swahili is “Wapi kituo cha gari moshi?” in German is “Wo ist der Bahnhof?” and in Amharic is “የባቡር ጣቢያው የት ነው?”

User: Thanks. Now identify the language in the following sentence “Ebee ka ọdụ ụgbọ igwe dị?”

Assistant: The sentence “Ebee ka ọdụ ụgbọ igwe dị?” is in Igbo.

User: What does it mean?

Assistant: The sentence “Ebee ka ọdụ ụgbọ igwe dị?” means “Where is the train station?” in English.

User: can you translate that sentence into Chinese and Hausa?

Assistant: The sentence “Ebee ka ọdụ ụgbọ igwe dị?” translates to “火车站在哪里” in Chinese and “wane ne tashar jirgin kasa?” in Hausa.

User: Thank you. How do you say “Goodbye and have a nice day” in Somali and Zulu?

Assistant: “Goodbye and have a nice day” in Somali is “Nabad salaaman tahay oo maalin wanaagsan” and in Zulu is “Sawubona futhi ube nosuku omuhle.”

User: danke schon :)

Assistant: “Thank you very much” in German is “Danke schön.”

User: correct, thank you and have a nice day!

Assistant: Have a nice day!

Figure 13: Multi-turn, conversational, cross-lingual translation and identification (TranslatePsy-AfriSLM-4B).